

E.3 / PAPER 2 / QUESTION 9

28 April 2022 / TZ1 / HL Paper 2 / original Q9 / 2222-6508

9. Potassium-40 ($^{40}_{19}\text{K}$) decays by two processes.

The first process is that of beta-minus (β^-) decay to form a calcium (Ca) nuclide.

- (a) Write down the equation for this decay.

[2]

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- (b) Potassium-40 decays by a second process to argon-40. This decay accounts for 11 % of the total decay of the potassium-40.

Rocks can be dated by measuring the quantity of argon-40 gas trapped in them. One rock sample contains $340\text{ }\mu\text{mol}$ of potassium-40 and $12\text{ }\mu\text{mol}$ of argon-40.

- (i) Show that the initial quantity of potassium-40 in the rock sample was about $450\text{ }\mu\text{mol}$.

[2]

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- (ii) The half-life of potassium-40 is 1.3×10^9 years. Estimate the age of the rock sample.

[3]

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- (c) Outline how the decay constant of potassium-40 was determined in the laboratory for a pure sample of the nuclide.

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References:

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